

Part V — Waves, Sound, and Light · Chapter 23

Vibrations and Oscillations

The repeating motions that underlie every wave in the physical world

A child on a swing moves back and forth, returning to the same position over and over. A guitar string, plucked and released, vibrates hundreds of times per second before the sound fades. The atoms in a crystal lattice vibrate around fixed positions at frequencies far beyond anything the human ear can detect. A quartz crystal in a watch oscillates so regularly that it keeps time to within seconds per year. In every case, the underlying physics is the same: a system displaced from equilibrium experiences a restoring force that drives it back, overshoots, and oscillates repeatedly. This chapter builds the conceptual and quantitative foundation for all wave phenomena by first understanding motion that repeats.

23.1 What Makes Motion Oscillatory

Most motion in nature does not repeat. A thrown ball follows a parabolic arc and lands. A car accelerates, cruises, and stops. These are one-way, non-repeating motions. Oscillatory motion is fundamentally different: the system moves, reverses, returns, and then repeats the same sequence indefinitely (or until energy is lost).

Three conditions produce oscillatory motion. First, the system must have an equilibrium position — a configuration toward which it naturally tends. Second, when displaced from equilibrium, the system must experience a restoring force directed back toward that equilibrium. Third, the system must have inertia: it carries momentum through the equilibrium position and overshoots to the other side, where the restoring force again pushes it back. It is the interplay of restoring force and inertia that sustains repetition.

Oscillation is not merely back-and-forth motion. It is a specific physical process in which energy cycles continuously between two forms: kinetic energy (associated with velocity through the equilibrium position) and potential energy (associated with displacement against the restoring force). Understanding this energy exchange is the key to understanding every oscillating system, from a pendulum to a quartz oscillator to the vibrating air column in a flute.

Key Point: Oscillation is local — the system moves back and forth around a fixed equilibrium. In the next chapter you will see that when oscillations propagate through

a medium, they create waves. Waves carry the oscillation from place to place. The oscillation itself stays local.

23.2 Periodic Motion and Repetition

Periodic motion is any motion that repeats itself at regular time intervals. Oscillatory motion is always periodic. Not all periodic motion is oscillatory in the strict sense — a ball rolling in a circular track repeats, but it does not oscillate around an equilibrium — but for the purposes of this chapter, we focus on the oscillatory case.

The key feature of periodic motion is the period T : the time required to complete one full cycle. After exactly one period, the system is back in the same position moving in the same direction with the same velocity as it started. Every measurable property of the system — position, velocity, acceleration, potential energy, kinetic energy — repeats with the same period T .

Periodicity is an idealization. Real oscillating systems lose energy to friction, air resistance, or internal damping, so the amplitude gradually decreases. But the period remains nearly constant even as energy is lost — a critically important property that makes oscillating systems useful as timekeepers. A pendulum clock keeps time not because the pendulum swings far, but because it swings regularly.

Historical Note: Galileo Galilei (1564–1642) is credited with discovering the isochronism of the pendulum — the property that a pendulum's period is nearly independent of its amplitude for small swings. Legend holds he observed a swinging lamp in the cathedral of Pisa and timed it against his pulse. Whether or not the story is accurate, his insight eventually led to the pendulum clock, which remained the world's most accurate timekeeper for nearly three centuries.

23.3 Equilibrium Position in Oscillating Systems

The equilibrium position is the configuration of the system where the net force is zero. It is the position the system naturally adopts when undisturbed. Crucially, the equilibrium position for an oscillating system is an unstable saddle in energy: the system will naturally return to it if displaced by a small amount (it is stable with respect to displacement), but the system will not stay there because it always arrives with kinetic energy it has gained during the return journey.

For a mass hanging on a spring, the equilibrium position is where the spring force exactly balances gravity. For a simple pendulum, it is the lowest point directly below the pivot. For a floating object, it is the level at which the buoyancy force equals the

object's weight. In each case, displacing the system from equilibrium creates a net force or torque directed back toward equilibrium.

A common misconception is that the system “wants to stay” at the equilibrium position. In fact, an undamped system passes through equilibrium at maximum speed and maximum kinetic energy — it cannot stop there unless energy is removed. The equilibrium is the point of minimum potential energy, but the total mechanical energy is conserved, so kinetic energy is at its maximum precisely when potential energy is at its minimum.

The Rule: At the equilibrium position: velocity is maximum, acceleration is zero, net force is zero, kinetic energy is maximum, potential energy is minimum. At maximum displacement: velocity is zero, acceleration is maximum, net force is maximum (toward equilibrium), kinetic energy is zero, potential energy is maximum.

23.4 Amplitude, Period, and Frequency

Four quantities fully describe the kinematics of a periodic oscillation: amplitude, period, frequency, and phase. At this level we focus on the first three.

Amplitude (A) is the maximum displacement from the equilibrium position. It measures how far the system swings or stretches in each direction. Amplitude is always a positive quantity. The system reaches $+A$ on one side and $-A$ on the other. Amplitude is determined entirely by initial conditions — how hard the system is displaced or pushed at the start. The amplitude of a spring oscillator tells you the maximum stretch or compression. The amplitude of a pendulum tells you the maximum angle from vertical.

Period (T) is the time for one complete cycle, measured in seconds. One complete cycle is one full excursion from the starting point, across to the far side, and back to the starting point moving in the original direction. For a pendulum, T is the time for a complete swing — left, right, and back to the starting side. For a mass on a spring, T is the time to go from maximum stretch to maximum compression and back.

Frequency (f) is the number of complete cycles per second, measured in hertz (Hz), where $1 \text{ Hz} = 1 \text{ cycle per second}$. A tuning fork vibrating at 440 Hz completes 440 full oscillations every second. Middle C on a piano is defined as 261.6 Hz. The human ear detects frequencies roughly from 20 Hz to 20 000 Hz. Ultrasound used in medical imaging operates at millions of hertz (MHz).

Quantity	Symbol	Unit	Meaning
Amplitude	A	m (or radians)	Maximum displacement from

			equilibrium
Period	T	s (seconds)	Time for one complete cycle
Frequency	f	Hz (hertz) = s^{-1}	Number of complete cycles per second
Angular frequency	ω	rad/s	Rate of phase change; $\omega = 2\pi f$

Table 23.1 Descriptors of periodic oscillation and their units.

23.5 Period–Frequency Relationship

Period and frequency are reciprocals of each other. If a system completes f cycles per second, each cycle takes $1/f$ seconds. Conversely, if each cycle takes T seconds, the system completes $1/T$ cycles per second.

$$f = 1/T \qquad T = 1/f$$

Frequency and period are reciprocals. One is fully determined by the other.

This reciprocal relationship is universal — it applies to every oscillating system without exception. A pendulum with a 2-second period has a frequency of 0.5 Hz. A guitar A-string vibrating at 440 Hz has a period of $1/440 \approx 0.00227 \text{ s} \approx 2.27 \text{ ms}$. A power-grid oscillation at 60 Hz has a period of $1/60 \approx 0.0167 \text{ s}$.

Example — Period to frequency A grandfather clock pendulum has a period of 2.00 s. Its frequency is $f = 1/T = 1/2.00 = 0.500 \text{ Hz}$. It completes exactly one half-swing (one tick) per second.

Example — Frequency to period Middle C on a piano has frequency 261.6 Hz. Its period is $T = 1/f = 1/261.6 = 3.82 \times 10^{-3} \text{ s} = 3.82 \text{ ms}$. Each full vibration takes less than 4 milliseconds.

Angular frequency ω (omega) is an alternative way to express frequency used in mathematical treatments of oscillation. It equals $2\pi f$, or equivalently $2\pi/T$, and has units of radians per second. The factor 2π converts cycles (each of which spans 2π radians of phase) to radians. Angular frequency appears naturally in the equations of simple harmonic motion.

$$\omega = 2\pi f = 2\pi/T$$

Angular frequency ω in radians per second. It measures how rapidly the phase of the oscillation advances.

23.6 Simple Harmonic Motion

Simple harmonic motion (SHM) is a specific, mathematically defined form of oscillatory motion in which the restoring force is directly proportional to displacement from equilibrium and always directed toward equilibrium. SHM is the simplest and most important type of oscillation in physics. Many real oscillating systems are exact SHM; many more are well approximated by SHM for small displacements.

In SHM, the position of the oscillating object as a function of time is a sinusoidal function — a sine or cosine curve. The object moves smoothly between its maximum positive and negative displacements, passing through equilibrium at maximum speed and slowing to momentary rest at the turning points.

$$x(t) = A \cos(\omega t + \varphi)$$

Position in SHM as a function of time. A = amplitude, ω = angular frequency, φ = phase constant (determines starting position).

The velocity in SHM is the derivative of position with respect to time, giving $v(t) = -A\omega \sin(\omega t + \varphi)$. The maximum speed (at equilibrium) is $v_{\text{max}} = A\omega = 2\pi fA$. The acceleration is the derivative of velocity: $a(t) = -A\omega^2 \cos(\omega t + \varphi) = -\omega^2 x(t)$. This last result is fundamental: in SHM, acceleration is always proportional to displacement and oppositely directed.

$$a = -\omega^2 x$$

In SHM, acceleration is proportional to displacement and directed oppositely. This is both the definition of SHM and a consequence of Hooke's Law applied to the restoring force.

The negative sign is crucial: when displacement is positive (to the right), acceleration is negative (to the left) — always restoring. When displacement is negative, acceleration is positive. This self-correcting behavior is what sustains the oscillation indefinitely in the absence of damping.

Bridge Note *The SHM equation $a = -\omega^2 x$ is a second-order differential equation whose solutions are sinusoidal functions. In university mathematics and physics, you will solve differential equations of this form and encounter complex exponential notation $e^{i\omega t}$. The conceptual framework built here translates directly.*

23.7 Restoring Forces

A restoring force is any force that acts to return a displaced system toward its equilibrium position. The defining property of a restoring force is that it is directed

toward equilibrium — it always opposes the displacement.

The simplest restoring force is proportional to displacement: $F = -kx$, where k is the stiffness constant of the system and x is the displacement from equilibrium. The negative sign signals that the force is always opposite to the displacement. This linear restoring force is Hooke's Law, and systems obeying it produce perfect SHM.

Many restoring forces are approximately linear for small displacements even if they are nonlinear for large ones. A pendulum's restoring force is $F = -mg \sin\theta$, which is nonlinear. But for small angles ($\theta \leq 15^\circ$), $\sin\theta \approx \theta$ (in radians), making the restoring force approximately linear and the motion approximately SHM. This small-angle approximation makes the pendulum solvable analytically and explains why pendulum clocks keep nearly perfect time as long as they do not swing too far.

Restoring forces arise in many physical systems. In a spring, it is the elastic force. In a pendulum, it is the component of gravity tangent to the arc. In a floating object, it is the excess or deficit of buoyancy when displaced vertically. In a molecular bond, it is the interatomic electromagnetic force. In each case, the same physics applies: displacement creates a restoring force, inertia causes overshoot, and oscillation results.

23.8 Mass–Spring Oscillators

The mass–spring system is the prototype oscillator of classical mechanics. It consists of a mass m attached to a spring with spring constant k . When displaced from equilibrium by a distance x , the spring exerts a restoring force $F = -kx$ (Hooke's Law). By Newton's second law, $F = ma$, giving $ma = -kx$, or $a = -(k/m)x$. Comparing this to the SHM condition $a = -\omega^2 x$, we immediately identify:

$$\omega = \sqrt{(k/m)}$$

$$T = 2\pi \sqrt{(m/k)}$$

$$f =$$

$$(1/2\pi) \sqrt{(k/m)}$$

Angular frequency, period, and frequency for a mass–spring oscillator. All three depend only on m and k , not on amplitude.

This result has three remarkable features. First, the period is independent of amplitude — a spring oscillator with a large displacement has exactly the same period as one with a small displacement (as long as the spring obeys Hooke's Law). Second, increasing the mass increases the period (heavier masses oscillate more slowly). Third, increasing the spring constant k decreases the period (stiffer springs oscillate more rapidly). These are all physically intuitive: a heavier mass has more inertia and responds more slowly; a stiffer spring has a stronger restoring force and pulls the mass back faster.

Example — Spring period calculation A 0.500 kg mass hangs from a spring with $k = 20.0 \text{ N/m}$. Period: $T = 2\pi\sqrt{(m/k)} = 2\pi\sqrt{(0.500/20.0)} = 2\pi\sqrt{(0.0250)} = 2\pi \times 0.158 = 0.993 \text{ s} \approx 1.00 \text{ s}$. Frequency: $f = 1/T = 1.00 \text{ Hz}$.

The mass–spring system can be oriented in any direction — horizontal on a frictionless surface, vertical (hanging), or even at an angle. For a vertical spring, gravity shifts the equilibrium position downward but does not change the period, because the oscillation occurs around the new (lower) equilibrium and gravity cancels in the equation of motion for displacement from equilibrium.

Key Point: For a vertical mass–spring system, the equilibrium position is where $mg = k\Delta x_0$ (Δx_0 is the static stretch of the spring due to gravity). When the mass oscillates, it oscillates around this new equilibrium, and the period formula $T = 2\pi\sqrt{(m/k)}$ still applies without any modification for gravity.

23.9 Pendulum Motion

A simple pendulum consists of a small mass (the bob) on a massless, inextensible string of length L , free to swing about a fixed pivot. When displaced to a small angle θ from vertical, the component of gravity tangent to the arc provides the restoring force: $F_{\text{tangent}} = -mg \sin\theta \approx -mg\theta$ for small angles. The equation of motion is formally identical to that of a spring oscillator, with mg/L playing the role of k and the angular displacement playing the role of x .

$$T = 2\pi\sqrt{(L/g)} \qquad f = (1/2\pi)\sqrt{(g/L)}$$

Period and frequency of a simple pendulum (small-angle approximation). L = string length, g = gravitational acceleration.

Three things are notable about this result. First, the period is independent of the mass of the bob — a heavy pendulum and a light pendulum of the same length oscillate with identical periods. This follows because both the restoring force and the inertia scale with mass, so mass cancels exactly. Second, the period depends on g : a pendulum on the Moon ($g_{\text{moon}} \approx 1.62 \text{ m/s}^2$) would swing about 2.5 times more slowly than the same pendulum on Earth. Third, the period increases with length — a longer pendulum swings more slowly.

Example — Pendulum period calculation A 1.00 m pendulum on Earth ($g = 9.81 \text{ m/s}^2$): $T = 2\pi\sqrt{(L/g)} = 2\pi\sqrt{(1.00/9.81)} = 2\pi \times 0.319 = 2.01 \text{ s}$. This is the basis of the “second pendulum”: a pendulum with $T = 2 \text{ s}$ has $L \approx 1 \text{ m}$, making one tick (half-period) exactly one second.

Pendulum motion deviates from the simple formula for large amplitudes. When the angle exceeds about 15° , the small-angle approximation fails and the period begins to

increase noticeably. At 30° amplitude, the true period is about 1.7% longer than the simple formula predicts. For pendulum clocks, this means the clock runs slightly slow when the pendulum swings wide — which is why early clockmakers designed mechanisms to keep the swing amplitude small and constant.

23.10 Energy in Oscillating Systems

One of the most powerful ways to analyze oscillatory motion is through energy. In an ideal (undamped) oscillator, mechanical energy is conserved — it does not increase or decrease but is continuously converted between kinetic and potential form.

For a mass–spring oscillator, the potential energy stored in the spring when it is displaced by x is $U = \frac{1}{2}kx^2$. The kinetic energy of the mass moving at speed v is $K = \frac{1}{2}mv^2$. At maximum displacement A , velocity is zero, so all energy is potential: $E = U_{\text{max}} = \frac{1}{2}kA^2$. At equilibrium ($x = 0$), all energy is kinetic: $E = K_{\text{max}} = \frac{1}{2}mv_{\text{max}}^2$. Since total energy is conserved:

$$E = \frac{1}{2}kA^2 = \frac{1}{2}kx^2 + \frac{1}{2}mv^2 = \text{constant}$$

Total mechanical energy of a mass–spring oscillator. It equals the potential energy at maximum displacement and the kinetic energy at equilibrium.

This energy equation lets us find the velocity at any position without needing the time-dependent equations. The maximum speed is $v_{\text{max}} = A\sqrt{k/m} = A\omega$. The total energy is proportional to the square of the amplitude: doubling the amplitude quadruples the energy.

The energy picture also provides insight into why amplitude does not affect period. The restoring force becomes stronger as amplitude increases (because k is constant but x is larger), and the maximum speed also increases. These two effects cancel exactly — the increased energy is “used up” at the same rate, keeping the period constant. This is a hallmark of true SHM with a linear restoring force.

Study Tip: The energy in an oscillating system is proportional to A^2 . Doubling the amplitude quadruples the energy. Halving the amplitude reduces the energy to one-quarter. This has direct practical implications: reducing vibration amplitude by a factor of 10 reduces the vibration energy by a factor of 100.

23.11 Damping and Energy Loss

All real oscillators lose energy over time. The mechanism may be friction between surfaces, air resistance, viscous drag in a fluid, or internal molecular rearrangement

within the material. Any process that dissipates mechanical energy into heat is a form of damping.

Damping has two effects. First, it reduces the amplitude of oscillation over time — the system swings less and less far with each cycle. Second, it slightly changes the period, though for lightly damped systems this effect is negligible. A lightly damped oscillator maintains its periodic character for many cycles before the amplitude becomes insignificantly small. A heavily damped oscillator returns to equilibrium without completing even one full oscillation.

The rate of amplitude decay depends on the damping strength. For a common type called viscous damping (where the damping force is proportional to velocity), the amplitude decreases exponentially with time. After each period, the amplitude is multiplied by a factor less than 1. The system traces out a pattern of oscillations that fit inside an exponentially shrinking envelope.

Damping is not always unwanted. Car shock absorbers are deliberately designed to be critically damped: they allow the suspension to absorb a bump and return to the normal position as quickly as possible without bouncing. A door closer, a galvanometer needle, and a bathroom scale all use deliberate damping to produce smooth, non-oscillatory behavior. The engineering question is always: how much damping is appropriate for the application?

Damping Type	Behavior	Physical Example
Undamped (ideal)	Amplitude constant indefinitely	Ideal pendulum (no friction)
Underdamped	Amplitude decreases slowly; many oscillations	Guitar string; clock pendulum
Critically damped	Returns to equilibrium fastest without oscillating	Car shock absorber; door closer
Overdamped	Returns to equilibrium slowly, no oscillation	Damper in heavy machinery

Table 23.2 *The four regimes of damped oscillation and their physical examples.*

23.12 Driven Oscillations

A driven oscillator is one subjected to a periodic external force in addition to its restoring force. The driving force supplies energy to the system, compensating for

damping and maintaining oscillation. If the driving force is absent, the oscillator gradually decays due to damping. With driving, the system can sustain indefinite oscillation.

The behavior of a driven oscillator depends critically on the relationship between the driving frequency f_d and the natural frequency f_0 (the frequency the system would oscillate at without driving). When these frequencies differ significantly, the system oscillates at the driving frequency with modest amplitude. When the driving frequency approaches the natural frequency, the situation changes dramatically.

The natural frequency (or resonant frequency) of a system is determined entirely by its physical properties: the mass and spring constant for a mass–spring system, the length and gravity for a pendulum. It is the frequency at which the system “wants” to oscillate. When a driver applies energy at this frequency, the system absorbs energy most efficiently, and the amplitude builds to a maximum.

The phase relationship between the driver and the oscillator also depends on the driving frequency. Below the natural frequency, the oscillator is nearly in phase with the driver. At the natural frequency, the oscillator lags the driver by exactly 90° . Above the natural frequency, the oscillator is nearly 180° out of phase. This phase behavior is exploited in electrical circuits (radio tuning, for example) to select specific frequencies.

23.13 Resonance (Introductory)

Resonance is the phenomenon in which a system oscillates with maximum amplitude when driven at its natural frequency. The driving force and the natural oscillation reinforce each other cycle after cycle, and amplitude grows until it is limited only by damping. In a lightly damped system, resonance can produce amplitudes vastly larger than those the driving force alone could cause.

Resonance is one of the most important phenomena in physics and engineering. It explains how a singer’s voice can shatter a crystal glass (the sound drives the glass at its natural frequency until the amplitude exceeds its breaking strength), how radio receivers select a single station from thousands (tuning adjusts the natural frequency of an electrical oscillator to match the broadcast frequency), how musical instruments amplify specific frequencies (the body of a guitar resonates at certain frequencies, boosting them), and how bridges can fail catastrophically when wind or traffic drives them near their natural frequencies.

*“Give me a lever long enough and a fulcrum on which to place it,
and I shall move the world. Give me a frequency precisely matched*

to the natural frequency of a structure, and I shall bring it down.” —

Adapted from engineering lore

The collapse of the Tacoma Narrows Bridge in 1940 is one of the most dramatic examples of resonance disaster. Wind-induced oscillations drove the bridge at a frequency near its torsional natural frequency, producing growing oscillations until the bridge twisted itself apart. Modern bridge design explicitly accounts for aeroelastic resonance by designing structures whose natural frequencies are far removed from any expected driving frequencies.

Beneficial resonance is equally important. MRI machines use nuclear magnetic resonance: hydrogen nuclei in the body precess at a natural frequency determined by the applied magnetic field. A radio-frequency pulse at exactly this frequency drives the nuclei into resonance, and the energy they emit as they relax is detected and used to form images. Laser cavities resonate at optical frequencies, producing coherent light. Musical resonance makes instruments expressive and responsive.

Historical Note: The Tacoma Narrows Bridge collapsed on November 7, 1940, just four months after opening. Film footage of the oscillating bridge — nicknamed “Galloping Gertie” — remains one of the most iconic videos in engineering history. The disaster led directly to modern understanding of aeroelastic flutter and transformed the design of suspension bridges worldwide.

Bridge Note *The mathematical treatment of resonance involves solving the driven harmonic oscillator differential equation, which introduces the concept of a frequency-dependent amplitude and Q -factor (quality factor). You will encounter this in university physics and electrical engineering, where it describes RLC circuits, mechanical vibrations, and optical resonators.*

23.14 Common Mistakes in Oscillation Analysis

Students learning about oscillatory motion make predictable errors. Awareness of these mistakes helps avoid them.

- Confusing period and frequency. Period is in seconds; frequency is in hertz (cycles per second). They are reciprocals. Never add or compare them directly without first converting one to the other.
- Assuming amplitude affects the period of a spring or pendulum. For an ideal spring and for a small-angle pendulum, the period is completely independent of amplitude. Only the natural frequency of the system (determined by m , k , L , g) sets the period.

- Forgetting that the pendulum formula requires radians and small angles. The formula $T = 2\pi\sqrt{L/g}$ is valid only for small amplitudes ($\theta \leq \text{about } 15^\circ$). For larger amplitudes, the true period is longer.
- Confusing the equilibrium position with the position of zero velocity. The equilibrium position is where the restoring force is zero, which is where velocity is maximum. The positions of zero velocity are at the turning points (maximum displacement).
- Forgetting that energy scales as A^2 , not A . Doubling the amplitude does not double the energy — it quadruples it.
- Believing that damping significantly changes the period. For light to moderate damping, the period is nearly unchanged. Damping primarily reduces amplitude, not period.
- Confusing natural frequency with driving frequency. The natural frequency is a property of the system. The driving frequency is chosen externally. Resonance occurs when they match.
- Treating resonance as always dangerous. Resonance can be beneficial (music, MRI, radio) or harmful (bridge failure, machinery vibration). The context determines whether resonance should be sought or avoided.

Chapter Summary

Oscillatory motion is repetitive motion around an equilibrium position, sustained by the interplay of a restoring force and inertia. It underlies all wave phenomena: waves occur when oscillation propagates through space.

- For oscillation to occur: an equilibrium position must exist, displacement must create a restoring force toward equilibrium, and the system must have inertia that carries it through equilibrium.
- Period T is the time for one complete cycle; frequency f is the number of cycles per second; they are reciprocals: $f = 1/T$.
- Angular frequency $\omega = 2\pi f = 2\pi/T$ measures the rate of phase advance in radians per second.
- Simple harmonic motion (SHM) occurs when the restoring force is proportional to displacement: $F = -kx$. Position varies sinusoidally: $x(t) = A \cos(\omega t + \phi)$.
- Mass–spring: $\omega = \sqrt{k/m}$; $T = 2\pi\sqrt{m/k}$. Period depends on m and k , not amplitude.
- Simple pendulum (small angle): $T = 2\pi\sqrt{L/g}$. Period depends on L and g , not mass or amplitude.

- Total energy of an oscillator equals $\frac{1}{2}kA^2$; it is proportional to A^2 and conserved in the absence of damping.
- Damping dissipates energy, reducing amplitude over time. Critically damped systems return to equilibrium fastest without oscillating.
- Driven oscillators are sustained by periodic external forces. Resonance occurs when the driving frequency matches the natural frequency, producing maximum amplitude.

Key Terms

oscillatory motion: Repetitive motion in which a system moves back and forth around an equilibrium position under the action of a restoring force.

periodic motion: Any motion that repeats itself at regular time intervals; characterized by a well-defined period T .

equilibrium position: The configuration of an oscillating system at which the net force is zero; the position around which oscillation occurs.

amplitude (A): The maximum displacement from equilibrium; determines the extent of the oscillation and is set by initial conditions.

period (T): The time required for one complete oscillation cycle; measured in seconds.

frequency (f): The number of complete oscillation cycles per second; measured in hertz (Hz); $f = 1/T$.

angular frequency (ω): The rate of phase change of the oscillation; $\omega = 2\pi f$; measured in radians per second.

simple harmonic motion (SHM): Oscillatory motion in which the restoring force is proportional to displacement; produces sinusoidal position vs. time; described by a $= -\omega^2 x$.

restoring force: A force directed toward the equilibrium position, whose magnitude is proportional to displacement for SHM: $F = -kx$.

spring constant (k): The stiffness of a spring; the proportionality constant in Hooke's Law $F = -kx$; measured in N/m.

natural frequency (f_0): The frequency at which a system oscillates freely when displaced; determined by the physical properties of the system.

damping: Any process that dissipates mechanical energy from an oscillating system, reducing amplitude over time.

driven oscillator: An oscillating system subjected to a periodic external driving force that compensates for energy loss and sustains oscillation.

resonance: The condition in which a system is driven at its natural frequency, producing maximum amplitude of oscillation.

isochronism: The property of an oscillator (e.g., a pendulum for small angles) in which the period is independent of amplitude.

Concept Review Questions

Answer in complete sentences. No calculations required.

1. What three conditions are necessary for a system to oscillate? Explain the role of each.
 2. Explain why an oscillating system passes through its equilibrium position at maximum speed rather than coming to rest there.
 3. A pendulum bob is pulled to one side and released. Describe the energy transformations that occur during one complete oscillation.
 4. Why is the period of a mass–spring oscillator independent of amplitude? Use energy reasoning in your answer.
 5. How does the period of a simple pendulum change if: (a) the mass is doubled, (b) the string length is quadrupled, (c) the pendulum is taken to the Moon?
 6. What is the difference between natural frequency and driving frequency? Under what condition does resonance occur?
 7. Explain what “critical damping” means and give a practical example of where it is deliberately engineered.
 8. Why is resonance sometimes beneficial and sometimes catastrophically harmful? Give one example of each.
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Oscillation Problems

Show all work. Include units throughout. Round appropriately.

9. An object oscillates with period $T = 0.400$ s. Find its frequency and angular frequency.

10. A tuning fork vibrates at 512 Hz. Find its period in milliseconds.
11. A mass–spring system has $m = 0.250$ kg and $k = 36.0$ N/m. Find (a) the angular frequency, (b) the period, and (c) the frequency.
12. A spring oscillator has $T = 1.20$ s. If the mass is increased by a factor of 4 (spring constant unchanged), what is the new period?
13. A spring oscillator has $f = 2.00$ Hz with a 0.500 kg mass. Find the spring constant k .
14. An oscillator has amplitude $A = 0.0800$ m and spring constant $k = 50.0$ N/m. Find (a) the total energy, (b) the maximum speed, and (c) the speed when $x = 0.0400$ m.
15. A mass on a spring oscillates with maximum speed 0.600 m/s and amplitude 0.0300 m. Find the angular frequency and spring constant if $m = 0.100$ kg.

Frequency and Period Exercises

16. The human heart at rest beats about 72 times per minute. Find the period (in seconds) and frequency (in Hz) of the heartbeat.
17. A metronome is set to 120 beats per minute. Find the period of each beat and the frequency in Hz.
18. Complete the table:

Period T (s)	Frequency f (Hz)	Angular Freq. ω (rad/s)
0.500	?	?
?	440	?
0.020	?	?
?	?	100

19. A clock pendulum completes 3600 full oscillations per hour. What is its period? Its frequency?
20. Radio station KPHP broadcasts at 98.7 MHz. Find the period of one oscillation of the broadcast carrier wave.

Spring and Pendulum Analysis

21. Find the length of a simple pendulum with a period of 1.00 s on Earth ($g = 9.81 \text{ m/s}^2$).
 22. A pendulum has length 0.625 m. Find its period on (a) Earth ($g = 9.81 \text{ m/s}^2$), (b) the Moon ($g = 1.62 \text{ m/s}^2$), and (c) Mars ($g = 3.72 \text{ m/s}^2$).
 23. A grandfather clock uses a 0.993 m pendulum on Earth. It is then used on a space station where $g_{\text{eff}} = 4.90 \text{ m/s}^2$. By how many minutes per hour will the clock gain or lose?
 24. A spring with $k = 80.0 \text{ N/m}$ and mass 0.320 kg oscillates with amplitude 5.00 cm. Find (a) the period, (b) the maximum kinetic energy, and (c) the position at which $KE = PE$.
 25. Two springs each with $k = 40.0 \text{ N/m}$ are connected to a 0.200 kg mass: (a) in series (effective $k_{\text{series}} = k/2$), and (b) in parallel (effective $k_{\text{parallel}} = 2k$). Find the period in each case.
 26. A mass-spring oscillator is constructed to have a period of exactly 0.500 s using a 0.400 kg mass. What spring constant is required?
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Resonance Concept Problems

27. A crystal wine glass has a natural frequency of 400 Hz. A singer sustains a note at 390 Hz near the glass. Will the glass resonate? Explain what would happen if the singer gradually raised the pitch to 400 Hz.
28. A child pumps a swing at a frequency of 0.5 Hz, which is the natural frequency of the swing. Explain why the amplitude builds with each push, even though each individual push is small.
29. A building is 20 stories tall and has a natural oscillation period of about 4 seconds. Earthquake P-waves arrive at 2 Hz. Are these waves likely to drive the building into resonance? Explain.
30. A car with worn shock absorbers tends to bounce at about 1.2 Hz on a rough road. At what road spacing would the bouncing become most severe if the car is traveling at 60 km/h? (Hint: find the road-feature period at that speed and spacing.)
31. Explain why radio tuning selects one station from thousands. What physical quantity of the receiver's circuit is adjusted, and to what value is it matched?

Chapter 23 Checkpoint

Before moving on to Chapter 24, confirm you can do each of the following without notes.

I can...	See Section
State the three conditions required for oscillatory motion	23.1
Define period, frequency, and amplitude and give their units	23.4
Convert between period, frequency, and angular frequency	23.5
Describe simple harmonic motion and state the condition $a = -\omega^2 x$	23.6
Explain what a restoring force is and give three examples	23.7
Calculate period and frequency for a mass–spring oscillator	23.8
Calculate period and frequency for a simple pendulum	23.9
Describe the energy exchange between kinetic and potential in an oscillator	23.10
Explain the four regimes of damping and give an example of each	23.11
Distinguish natural frequency from driving frequency	23.12
Define resonance and explain one beneficial and one harmful example	23.13
Identify and avoid the most common errors in oscillation problems	23.14

Continue to Chapter 24: Wave Properties →